Lab Assignment 4: Frame-wise Analysis of Speech Signal

Title: To perform frame-wise analysis of a speech signal and study the behaviour of time-domain features for voiced and unvoiced speech.

1. Introduction

Short-time processing is a fundamental technique in speech analysis because speech is a non-stationary signal whose properties change over time. By segmenting the signal into small, quasi-stationary frames (typically 20-30ms), we can extract time-domain features that help distinguish between different types of speech sounds, specifically voiced and unvoiced segments.

2. Methodology & Experimental Setup

2.1 Dataset

- **Sampling Frequency:** 16,000 Hz (16 kHz).
- **Segment Analyzed:** First 20 seconds of the speech sample.

2.2 Framing Parameters

Parameter	Value
Frame Length	25 ms (400 samples)
Frame Shift	10 ms (160 samples)
Total Frames	Calculated based on signal duration

2.3 Window Functions

Two window functions were utilized for feature extraction to compare their effects:

1. **Hamming Window:** Designed to minimize the maximum side-lobe level.
2. **Rectangular Window:** Provides the narrowest main-lobe but higher side-lobes.

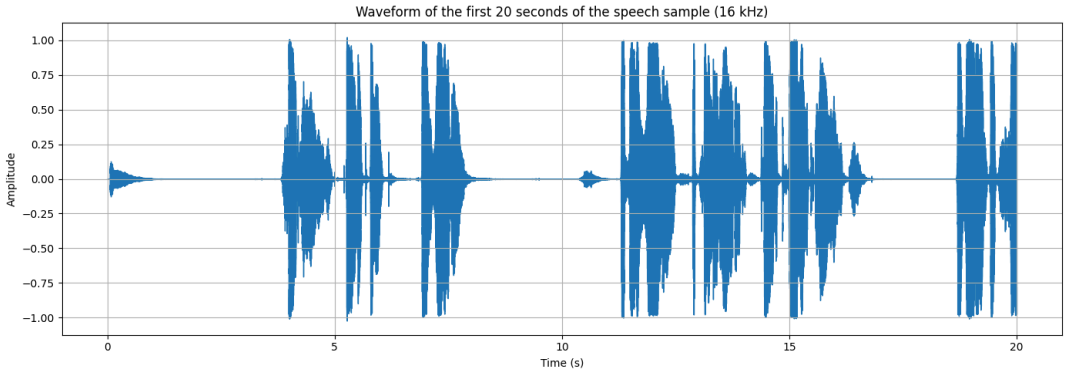
3. Task 1: Short-Time Feature Extraction

3.1 Time-Domain Features Extracted

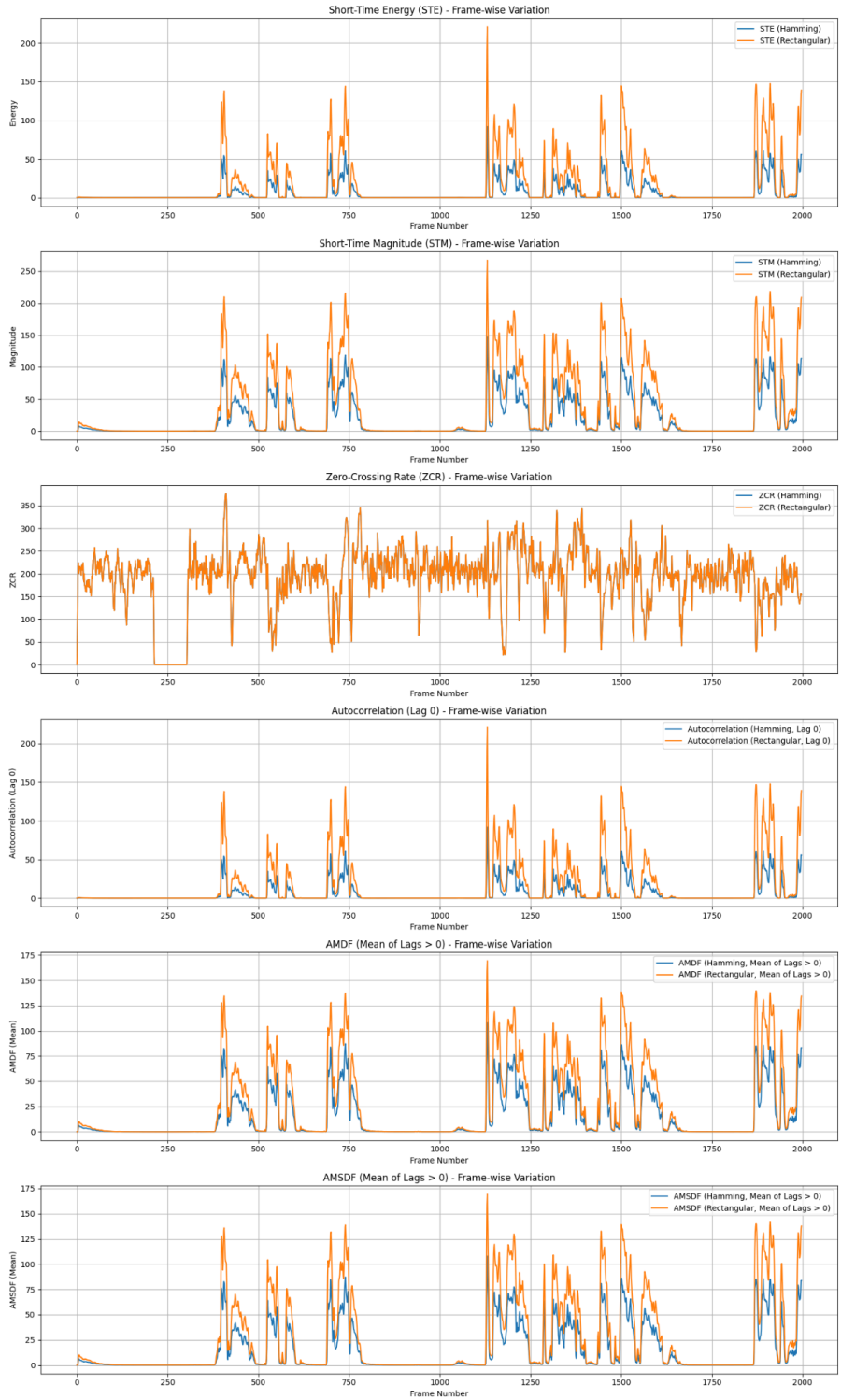
- **Short-Time Energy (STE):** Measures the signal intensity.
- **Short-Time Magnitude (STM):** Similar to energy but less sensitive to high-amplitude peaks.
- **Zero-Crossing Rate (ZCR):** Measures the frequency of sign changes; a proxy for spectral content.
- **Autocorrelation:** Measures self-similarity at different lags.
- **AMDF (Average Magnitude Difference Function):** A difference-based periodicity measure.
- **AMSDF (Average Magnitude Squared Difference Function):** Similar to AMDF but uses squared differences.

3.2 Visual Analysis of Waveform

Description: The waveform shows distinct regions of high-amplitude periodic oscillations (voiced) and low-amplitude noise-like bursts (unvoiced).



3.3 Feature Comparison: Hamming vs. Rectangular



Observation on Feature Smoothness:

Short Inference: Hamming vs Rectangular Window (All Features)

1. STE & STM: Both windows clearly separate voiced (high values) and unvoiced (low values) speech. The rectangular window produces higher and sharper peaks, while the Hamming window gives smoother and slightly reduced magnitudes.

2. ZCR: Voiced segments show low ZCR and unvoiced segments show high ZCR for both windows. Very little difference between Hamming and rectangular, making ZCR less sensitive to window type.

3. Autocorrelation (Lag 0): Follows the same pattern as STE. The rectangular window gives higher peaks, whereas the Hamming window smooths the variations.

4. AMDF & AMSDF: Higher in voiced regions and lower in unvoiced regions for both windows. The rectangular window shows stronger fluctuations, while the Hamming window provides smoother feature curves.

Both windows effectively distinguish voiced and unvoiced speech. The rectangular window emphasizes peak amplitudes, while the Hamming window provides smoother and more stable feature representation.

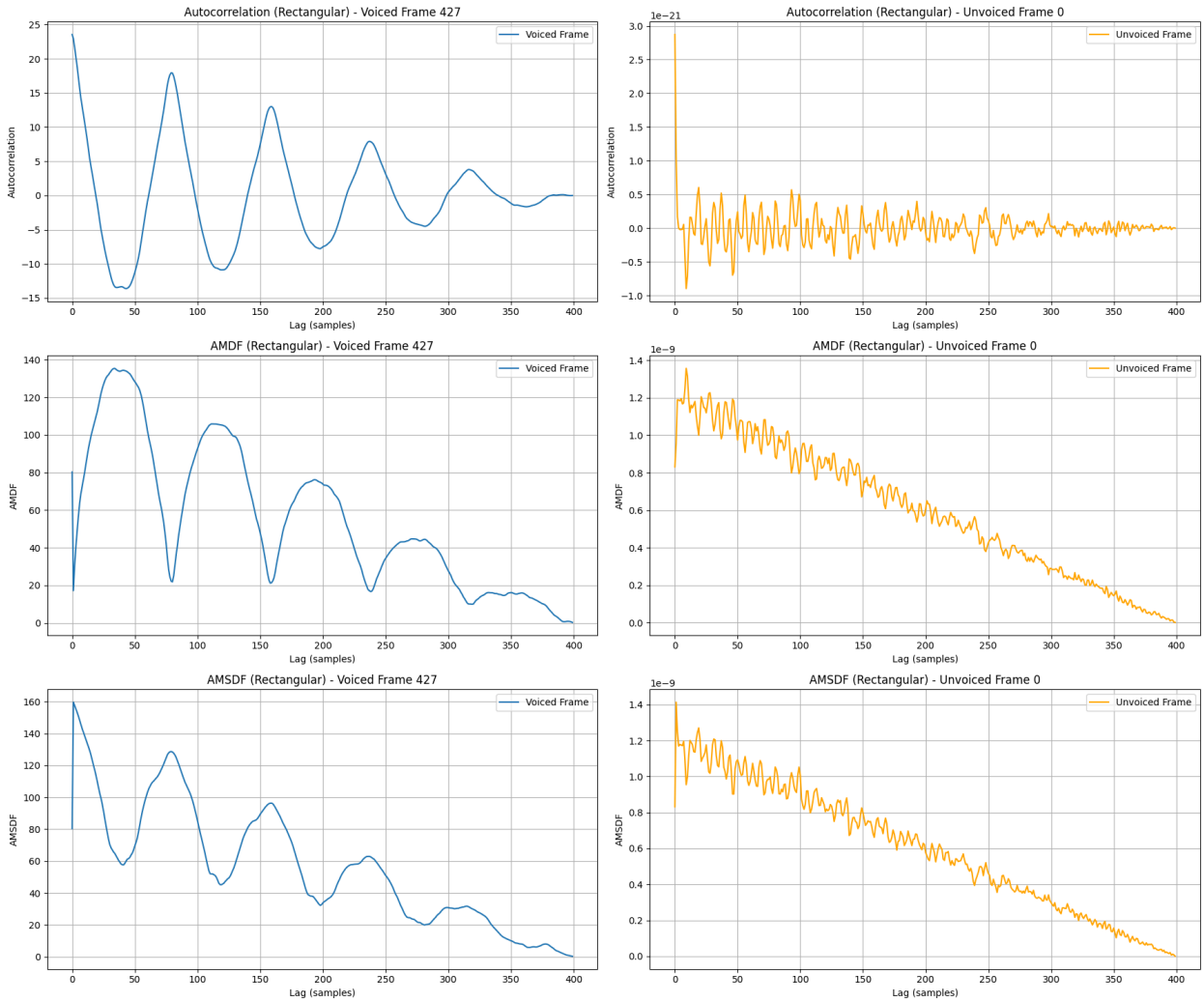
Window Choice:

The Hamming window produces smoother feature curves with lower standard deviation due to edge tapering. This reduces abrupt fluctuations and provides more stable feature representation across frames.

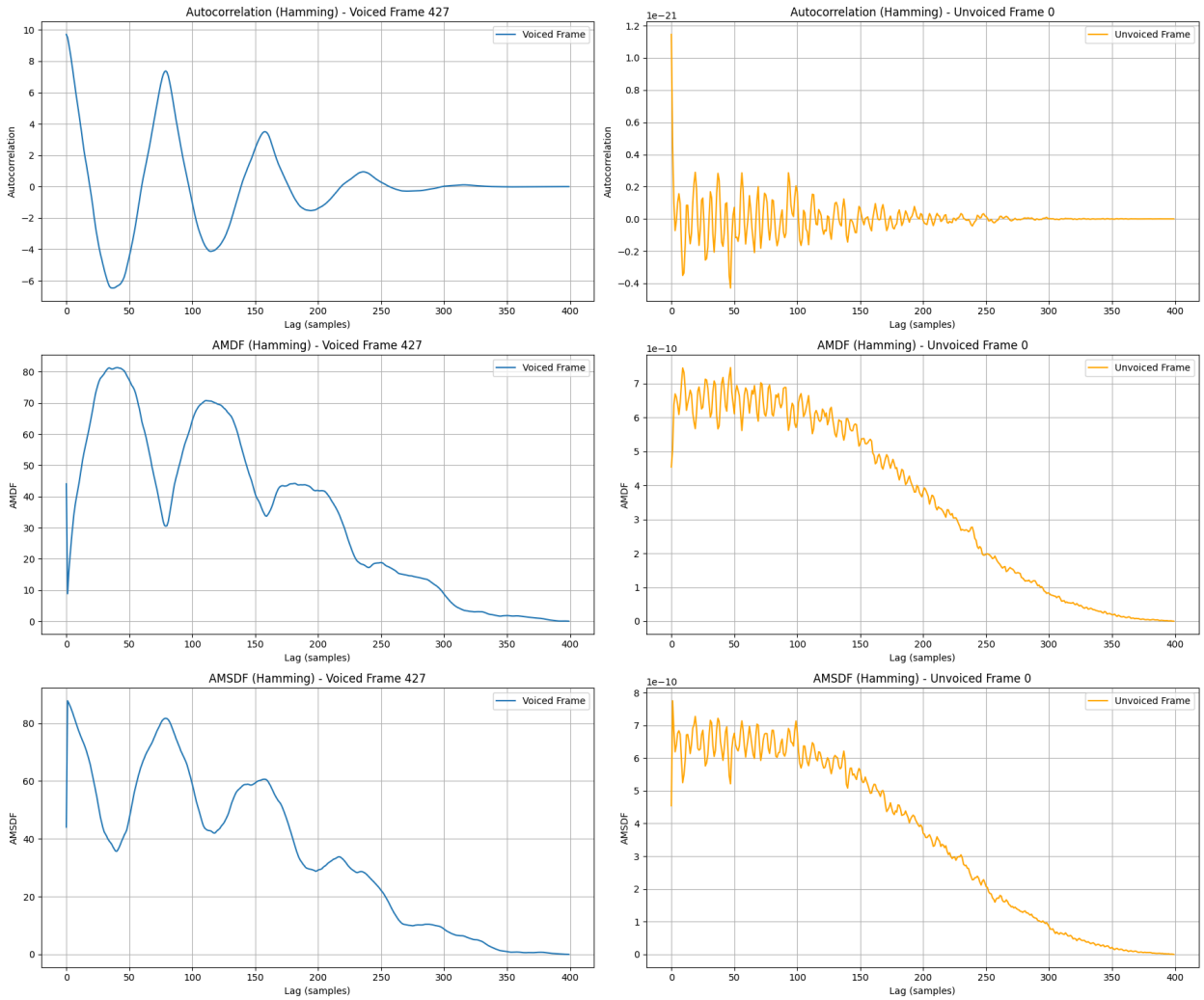
The Rectangular window results in higher magnitudes and larger standard deviations, leading to sharper peaks and more fluctuations.

4. Task 2 & 3: Periodicity Analysis and Pitch Estimation

4.1 Periodic (Voiced) vs. Aperiodic (Unvoiced)



Comparison of periodicity in case of voiced and unvoiced speech signals in Rectangular window



Comparison of periodicity in case of voiced and unvoiced speech signals in Hamming window

(b) Reflection of Periodicity in the Functions

For voiced (periodic) frames:

1. Autocorrelation: Clear repeating peaks occur at multiples of the pitch period. These peaks indicate strong similarity between the signal and its delayed versions, confirming periodicity.

2. AMDF / AMSDF: Distinct minima appear at lags corresponding to the pitch period. Since the signal repeats, the difference between the original and delayed signal becomes small, producing valleys.

For unvoiced (aperiodic) frames:

1. Autocorrelation: No clear secondary peaks beyond lag 0. The function quickly decays and fluctuates randomly, indicating lack of periodic structure.

2. AMDF / AMSDF: No distinct periodic minima. The curves are irregular and gradually decay, reflecting noise-like behavior.

(c) Distinction Between Periodic (Voiced) and Aperiodic (Unvoiced) Frames

1. Periodic (Voiced) Frames:

- Strong, regularly spaced peaks in autocorrelation.
- Clear, repeating minima in AMDF/AMSDF.
- Structured and smooth oscillatory patterns.
- Indicate presence of pitch.

2. Aperiodic (Unvoiced) Frames:

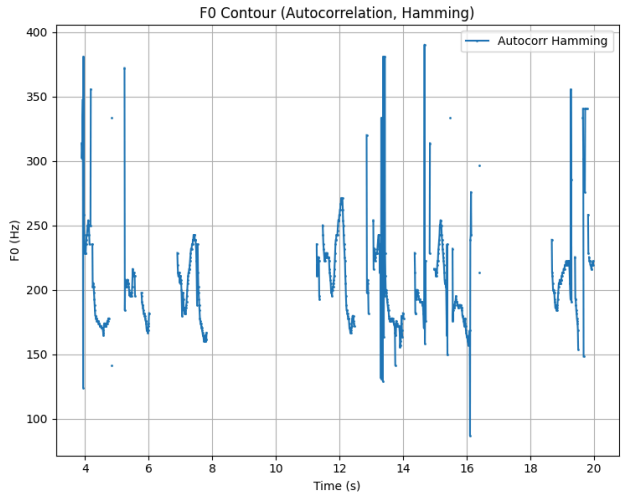
- Only a dominant peak at lag 0 in autocorrelation.
- No consistent minima in AMDF/AMSDF.
- Irregular, noise-like fluctuations.
- No pitch structure.

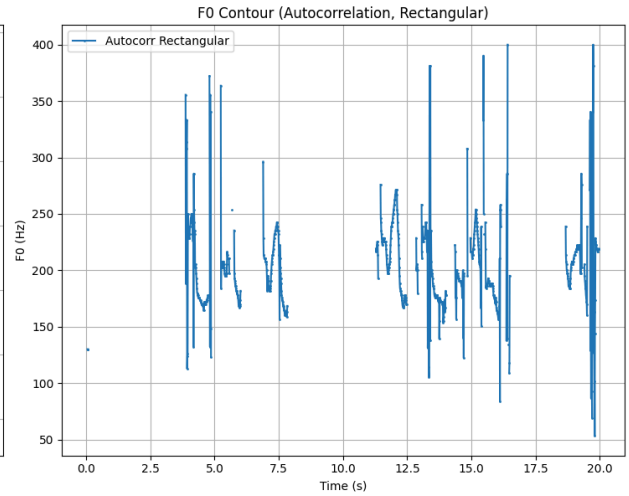
Conclusion:

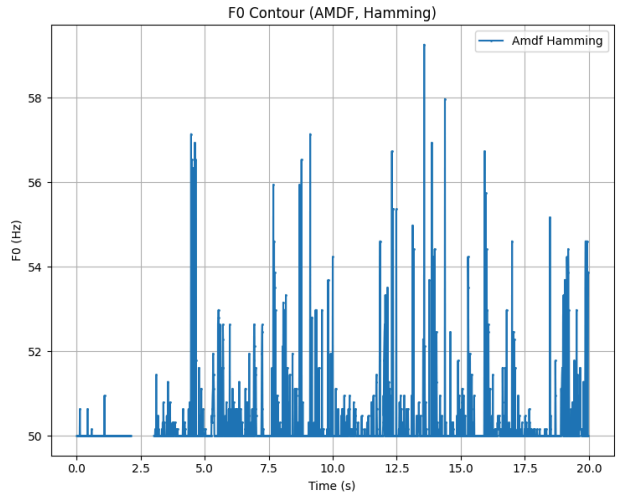
Periodicity is characterized by repeating peaks (autocorrelation) and repeating minima (AMDF/AMSDF). Their absence indicates unvoiced, noise-like speech.

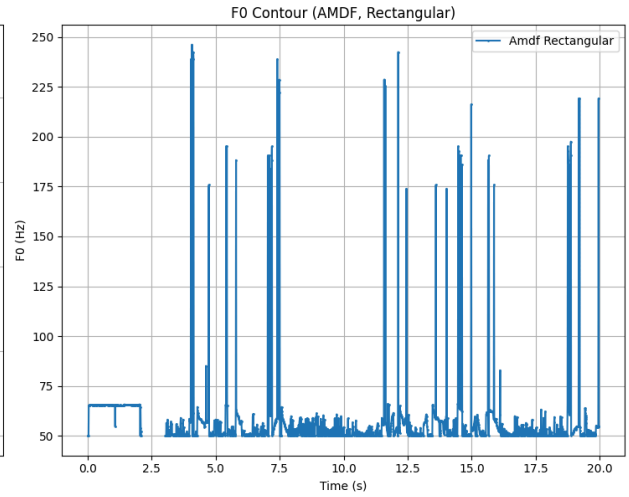
4.2 Pitch Frequency (F0) Results

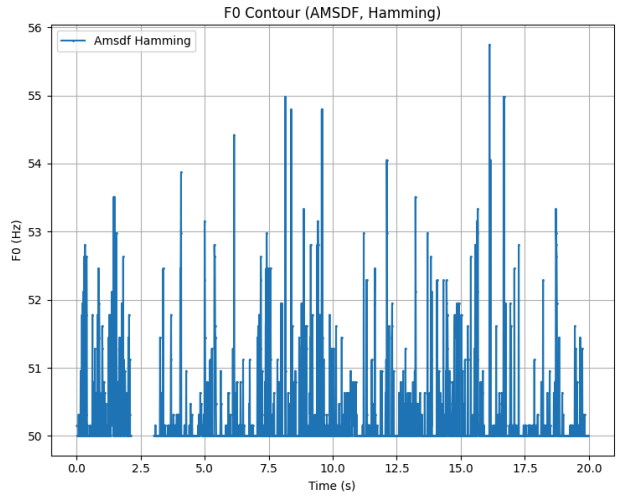
Based on the periodicity analysis, the fundamental frequency was estimated by finding the lag (k) corresponding to the first prominent peak (Autocorr) or minimum (AMDF).

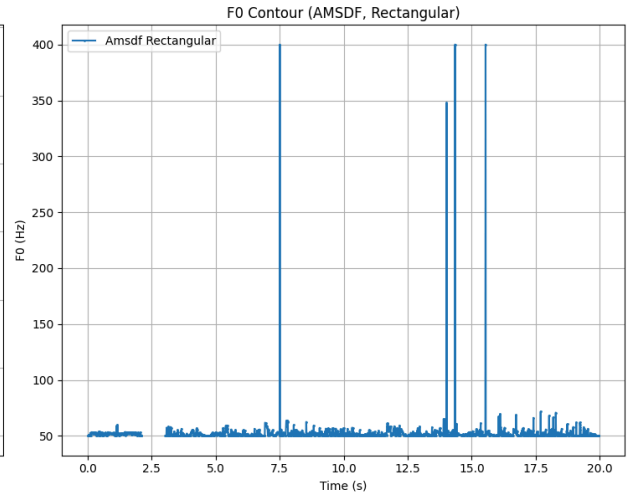












From the F0 contour plots:

The dominant pitch region observed in the autocorrelation method lies between 170–220 Hz.

Taking the central tendency, the estimated pitch frequency is: ≈200 Hz

AMDF and AMSDF (Hamming) show clustered values around 50–60 Hz, which likely correspond to a subharmonic (larger lag selection) rather than the true fundamental frequency to reduce discontinuities, suppress artificial high-frequency components.

Rectangular window results show additional spikes (250–400 Hz) due to unstable lag selection.

Inference:

The true pitch frequency of the voiced speech is approximately ~200 Hz, which falls within the normal human pitch range. The wide variations observed in some contours are caused by octave/subharmonic errors and peak selection instability, not actual rapid changes in pitch.

5. Conclusion

Feature Effectiveness:
STE, STM, ZCR, Autocorrelation, AMDF, and AMSDF effectively distinguished voiced and unvoiced speech. Voiced segments showed higher energy-based features and lower ZCR, while unvoiced segments exhibited the opposite. Autocorrelation peaks and AMDF/AMSDF minima clearly reflected periodicity in voiced frames.

Impact of Windowing:
The Hamming window produced smoother and more stable feature curves due to its tapering effect, reducing edge artifacts. The Rectangular window preserved higher magnitudes but resulted in noisier and more fluctuating contours. ZCR remained largely unaffected by window choice.

Pitch Estimation:
Autocorrelation identified the dominant pitch region around ~200 Hz, corresponding to the true fundamental frequency. AMDF and AMSDF produced smoother contours (especially with Hamming), but frequently selected subharmonic lags (~50–60 Hz), indicating octave errors. Thus, while AMDF/AMSDF were more stable, Autocorrelation provided a pitch estimate closer to the true F0.

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